# Codebase Copilot — Complete Technical Deep Dive

## Interview Preparation Guide

**Author:** Built from scratch using Python, LangGraph, Qdrant, FastAPI, HuggingFace, and Google Gemini
**Purpose:** A production-grade, multi-agent AI system that ingests any codebase and answers questions, generates reviews, writes tests, and produces documentation — with zero hallucinations.

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# 1. Project Overview & Problem Statement

### The Core Problem

Large codebases are notoriously hard to navigate. Developers spend a significant portion of their workday just figuring out *what existing code does*, not writing new code.

**Standard LLMs (like ChatGPT) fail here for three reasons:**

1. **Training Cutoff**: Your private codebase wasn't in their training data.
2. **Hallucinations**: If they don't know, they make things up. This is dangerous for code generation.
3. **Context Window Limits**: You can't paste 1 million lines of code into a single prompt.

### The Solution: Retrieval-Augmented Generation (RAG) + Multi-Agent AI

Instead of relying on memorized knowledge, our system:

1. **Ingests** the codebase into a searchable database *once* (offline).
2. **Retrieves** only the most relevant code snippets at query time (just like an "open book exam").

3. **Routes** the query to the best specialized AI agent for that task.
4. **Generates** a verified, citation-backed answer using the retrieved context.

### What Makes This Advanced (Not a Basic Tutorial)

- Uses **AST-level chunking** instead of dumb text splitting.
- Uses **Hybrid Search with RRF** (not just vector search).
- Uses **Cross-Encoder Reranking** for precision.
- Uses **Dependency Graph Expansion** for multi-hop reasoning.
- Uses **LangGraph State Machines** for agent orchestration.
- Uses **Self-Verification Loops** to eliminate hallucinations.
- Uses **Cross-Provider Fallback** with exponential backoff for production reliability.

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## 2. Technology Stack

| Technology | Role | Why This Specifically |
|---|---|---|
| **Python 3.12** | Core language | Dominant in AI/ML ecosystem |
| **Tree-Sitter** | AST Parser | Industry-standard (used by GitHub, Neovim). Language-agnostic structural parsing |
| **Qdrant** | Vector Database | Blazing fast, Rust-based, supports local file mode & Docker |
| **HuggingFace sentence-transformers** | Embedding model | `all-MiniLM-L6-v2` is the best quality/speed tradeoff for code |
| **BM25Okapi (rank-bm25)** | Sparse keyword search | Proven IR algorithm, handles exact symbol matching |
| **CrossEncoder (sentence-transformers)** | Reranker | Re-scores candidates with query context for maximum precision |
| **LangChain** | LLM abstraction | Unified interface across Gemini/Anthropic, chain composition |
| **LangGraph** | Agent orchestration | State-machine graph for deterministic multi-agent routing |
| **Google Gemini** | Primary LLM | Gemini 3.7 Flash (complex tasks), Gemini 3.5 Flash (routing) |
| **Anthropic Claude** | Fallback LLM | `claude-3-5-sonnet` as primary fallback, `claude-3-haiku` as fast fallback |
| **FastAPI** | Web API backend | Async, auto-generates OpenAPI docs, high performance |
| **Qdrant** | Vector DB | Local file persistence without needing Docker |
| **RAGAS** | Evaluation framework | Industry-standard LLM evaluation metrics |

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## 3. Phase 1: The Ingestion Pipeline

This phase runs **once**, offline, to prepare the codebase for AI-powered querying.

### 3.1 Overall Ingestion Flow

```
Source Code Files
       |
       v
[ingestion/chunker.py]
  - Tree-Sitter AST Parser
  - Extracts: functions, classes
  - Tags: file path, line numbers, calls made
       |
       v
[ingestion/graph_builder.py]
  - Builds Caller/Callee Dependency Graph
  - Saves to dependency_graph.json
       |
       +------------------+
       |                  |
       v                  v
[HuggingFace Embed]  [BM25Okapi Index]
 Dense Vectors        Sparse Index
       |                  |
       v                  v
  [Qdrant DB]        [bm25_index.pkl]
  (qdrant_data/)
```

## 3.2 Deep Dive: AST Chunking (`ingestion/chunker.py`)

**Why not split by newlines or fixed character limits?**
A naive chunker might cut a function in half. The AI then sees an incomplete snippet, cannot understand it, and will hallucinate. This is the #1 cause of poor RAG quality.

**Our Approach: Tree-Sitter**
Tree-Sitter parses source code into an **Abstract Syntax Tree (AST)** — a tree data structure representing the *syntactic structure* of the code. We traverse this tree and extract only complete, semantically cohesive blocks.

**Code Walkthrough:**

```
# chunker.py — Key logic
PY_LANGUAGE = Language(tspython.language())  # Load Python grammar

class CodeChunker:
    def chunk_file(self, file_path: str):
        tree = self.parser.parse(bytes(code, "utf8"))  # Build the AST

        def traverse(node):
            # Only extract complete function & class definitions
            if node.type in ['function_definition', 'class_definition']
                name = ... # get the identifier name
                calls = ... # find all function calls INSIDE this block

                chunks.append({
                    "file_path": file_path,
                    "symbol_name": name,      # e.g., "multiply"
                    "start_line": node.start_point[0],
```

```
                    "end_line": node.end_point[0],
                    "content": code[node.start_byte:node.end_byte],
                    "calls": list(set(calls))  # dependency edges
                })

            for child in node.children:
                traverse(child)  # recurse into nested classes/methods
```

**What each chunk looks like:**

```
{
  "file_path": "dummy_repo/math_utils.py",
  "symbol_name": "multiply",
  "type": "function_definition",
  "start_line": 1,
  "end_line": 3,
  "content": "def multiply(a, b):\n    return a * b",
  "calls": []
}
```

## 3.3 Deep Dive: Dependency Graph (`ingestion/graph_builder.py`)

**Why do we need a dependency graph?**
Imagine a user asks "Why does the `process_order` function fail?" The relevant code isn't just `process_order` itself — it also includes `validate_payment`, `update_inventory`, and `send_confirmation`, which it calls. Standard RAG would miss these.

**Our Approach: Bidirectional Call Graph**
During chunking, we extract every function call made *inside* each code block. We then build a graph where:

- `function_A.calls = [function_B, function_C]`
- `function_B.called_by = [function_A]`

```
# graph_builder.py — Key logic
class DependencyGraph:
    def add_chunk(self, chunk):
        symbol = chunk['symbol_name']    # e.g., "process_order"
        calls = chunk['calls']            # e.g., ["validate_payment",

        for call in calls:
            self.graph[symbol]["calls"].add(call)
            self.graph[call]["called_by"].add(symbol)  # bidirectional!
```

**Output (`dependency_graph.json`):**

```
{
  "process_order": {"calls": ["validate_payment", "send_email"], "calle
  "validate_payment": {"calls": [], "called_by": ["process_order"]}
}
```

## 3.4 Deep Dive: Vector & BM25 Indexing (`ingestion/indexer.py`)

Two separate indexes are built simultaneously:

**Dense Vector Index (Qdrant):**

- Each chunk's `content` is converted into a 384-dimensional float vector using `HuggingFaceEmbeddings(model_name="all-MiniLM-L6-v2")`.
- This model was pre-trained on 1 billion sentence pairs and understands semantic similarity.
- Vectors are stored in Qdrant with `COSINE` distance metric.

**Sparse BM25 Index:**

- The corpus is tokenized: `"def multiply(a, b)".lower().split()` → `["def", "multiply", "a,", "b)"]`.
- `BM25Okapi` builds an inverted index with TF-IDF-like scoring.
- The entire index is pickled to `bm25_index.pkl` for fast reloading.

```
# indexer.py — Both indexing strategies
# Dense (Semantic)
vector = self.embeddings.embed_query(chunk["content"])
points.append(PointStruct(id=i, vector=vector, payload={...}))
self.qdrant.upsert(collection_name=..., points=points)

# Sparse (Keyword)
tokenized_corpus = [chunk["content"].lower().split() for chunk in all_c
self.bm25 = BM25Okapi(tokenized_corpus)
pickle.dump({"bm25": self.bm25, "chunks": all_chunks}, f)
```

---

# 4. Phase 2: The Retrieval Layer

Every time a user submits a query, this three-stage retrieval pipeline runs to find the best possible context before handing it to the LLM.

## 4.1 Three-Stage Retrieval Flow

```
User Query
    |
    v
[Stage 1: Hybrid Search — retrieval/hybrid_search.py]
    |-- Dense Vector Search (Qdrant) --> Top 30 semantic matches
    |-- Sparse BM25 Search           --> Top 30 keyword matches
    |-- Reciprocal Rank Fusion (RRF) --> Merged Top 30 list
    |
    v
[Stage 2: Cross-Encoder Reranking — retrieval/reranker.py]
    |-- Scores each chunk against query using a Cross-Encoder model
    |-- Returns Top 5 most relevant chunks
    |
    v
[Stage 3: Multi-Hop Expansion — retrieval/multi_hop.py]
    |-- Reads dependency_graph.json
    |-- For each Top 5 chunk, adds its callers and callees
    |-- Returns 4-12 final chunks (enriched context)
    |
```

```
    v
Context String fed into LLM
```

## 4.2 Deep Dive: Hybrid Search + RRF (`retrieval/hybrid_search.py`)

**The Problem with Single-Strategy Search:**

- **Dense-only**: Can't handle exact symbol lookups like `MAX_RETRY_COUNT`.
- **BM25-only**: Can't handle conceptual queries like "function that validates user input".

**Our Solution: Run Both, Merge with RRF**

The **Reciprocal Rank Fusion** formula:

```
RRF_score(chunk) = Σ  1 / (k + rank_in_list)
```

Where `k=60` (a constant to reduce the impact of high ranks) and `rank` is the chunk's position in each result list. A chunk that ranks #1 in both lists scores `1/61 + 1/61 ≈ 0.033`, while a chunk ranking #30 in both scores `1/90 + 1/90 ≈ 0.022`.

**Code Walkthrough:**

```
# hybrid_search.py — The RRF core
def search(self, query: str, top_k: int = 30):
    # 1. Dense Search via Qdrant
    vector = self.embeddings.embed_query(query)
    dense_results = self.qdrant.query_points(collection_name=..., query
    dense_ranked = [f"{hit.payload['file_path']}::{hit.payload['symbol_
                    for hit in dense_results.points]

    # 2. Sparse BM25 Search
    tokenized_query = query.lower().split()
    sparse_scores = self.bm25.get_scores(tokenized_query)
    sparse_top_indices = sorted(range(len(sparse_scores)),
                                key=lambda i: sparse_scores[i], revers
    sparse_ranked = [f"{chunks[i]['file_path']}::{chunks[i]['symbol_nam
                     for i in sparse_top_indices]

    # 3. RRF Merge
    rrf_scores = {}
    for rank, chunk_id in enumerate(dense_ranked):
        rrf_scores[chunk_id] = rrf_scores.get(chunk_id, 0) + 1.0 / (60
    for rank, chunk_id in enumerate(sparse_ranked):
        rrf_scores[chunk_id] = rrf_scores.get(chunk_id, 0) + 1.0 / (60

    # 4. Sort and return top chunks
    sorted_chunks = sorted(rrf_scores.items(), key=lambda x: x[1], reve
```

## 4.3 Deep Dive: Cross-Encoder Reranking (`retrieval/reranker.py`)

After the initial retrieval we have 30 candidates. The problem: **biencoder models** (like the HuggingFace embedding model) encode the query and the document *separately*, so they miss fine-grained semantic interactions.

**Cross-Encoders** process the query and document *together* in one model pass, producing a much more accurate relevance score (at the cost of speed — which is why we only rerank 30 candidates, not all chunks).

```
# reranker.py
class Reranker:
    def __init__(self):
        # ms-marco is trained specifically for passage ranking
        self.model = CrossEncoder("cross-encoder/ms-marco-MiniLM-L-6-v2

    def rerank(self, query: str, chunks: List[Dict], top_k: int = 5):
        # Feed each (query, document) pair to the cross-encoder
        model_inputs = [[query, chunk["content"]] for chunk in chunks]
        scores = self.model.predict(model_inputs)    # One score per pa

        # Sort by score and return top 5
        scored_chunks = list(zip(chunks, scores))
        scored_chunks.sort(key=lambda x: x[1], reverse=True)
        return [chunk for chunk, score in scored_chunks[:top_k]]
```

### 4.4 Deep Dive: Multi-Hop Expansion (`retrieval/multi_hop.py`)

Takes the top-5 reranked results and enriches them with dependency context.

```
# multi_hop.py
def expand(self, chunks: List[Dict], all_chunks: List[Dict]) -> List[Di
    chunk_map = {c['symbol_name']: c for c in all_chunks}
    expanded = list(chunks)
    seen = set([c['symbol_name'] for c in chunks])

    for chunk in chunks:
        symbol = chunk['symbol_name']
        if symbol in self.graph:
            # Pull in functions this chunk calls
            for callee in self.graph[symbol].get("calls", []):
                if callee in chunk_map and callee not in seen:
                    expanded.append(chunk_map[callee])
                    seen.add(callee)
            # Pull in functions that call this chunk
            for caller in self.graph[symbol].get("called_by", []):
                if caller in chunk_map and caller not in seen:
                    expanded.append(chunk_map[caller])
                    seen.add(caller)
    return expanded
```

---

## 5. Phase 3: Multi-Agent Orchestration (LangGraph)

### 5.1 Why Multiple Agents Instead of One Big Prompt?

A single prompt like *"You are a code assistant. You can answer questions, review PRs, write tests, and generate docs"* leads to:

- **Attention dilution**: The model tries to satisfy all constraints simultaneously.
- **Lower quality outputs**: The QA system prompt is different from the Test Writing system prompt.
- **No specialization**: A Senior SDET and a Technical Writer have fundamentally different cognitive modes.

**Multi-agent** systems solve this by assigning each task a dedicated, laser-focused agent with a highly specific system prompt.

## 5.2 LangGraph State Machine Architecture

```
User Prompt enters AppState
         |
         v
  [Supervisor Node]
  classify_intent()
  Uses: Gemini 3.5 Flash + routing_prompt
  Outputs: one of "qa" | "review" | "test" | "docs"
         |
    +---------+---------+---------+
    |         |         |         |
    v         v         v         v
  [QA]   [Review]  [Test]    [Docs]
  Node     Node     Node      Node
    |         |         |         |
    +---------+---------+---------+
                   |
                   v
              [END Node]
          Returns final_output
```

**The AppState TypedDict (Shared Memory):**

```
class AppState(TypedDict):
    input_text: str    # The user's raw prompt
    intent: str        # Set by Supervisor: "qa", "review", "test", "do
    final_output: str  # Set by the worker agent: the generated text
```

## 5.3 Code Walkthrough: Building the Graph (`agents/supervisor.py`)

```
def _build_graph(self):
    workflow = StateGraph(AppState)

    # Register all nodes
    workflow.add_node("supervisor", self.classify_intent)
    workflow.add_node("qa", self.run_qa)
    workflow.add_node("review", self.run_review)
    workflow.add_node("test", self.run_test)
    workflow.add_node("docs", self.run_docs)

    workflow.set_entry_point("supervisor")

    # Conditional routing: reads state["intent"] to decide next node
```

```
    workflow.add_conditional_edges(
        "supervisor",
        lambda state: state["intent"],
        {"qa": "qa", "review": "review", "test": "test", "docs": "docs"
    )

    # All worker nodes end the graph
    workflow.add_edge("qa", END)
    workflow.add_edge("review", END)
    workflow.add_edge("test", END)
    workflow.add_edge("docs", END)

    return workflow.compile()

def classify_intent(self, state: AppState):
    # LangChain Expression Language (LCEL) chain
    chain = self.routing_prompt | self.llm | StrOutputParser()
    intent = chain.invoke({"input_text": state["input_text"]}).strip().

    if intent not in ["qa", "review", "test", "docs"]:
        intent = "qa"  # Graceful fallback

    return {"intent": intent}
```

---

# 6. Phase 4: The Four Specialized Agents

## 6.1 QA Agent (`agents/qa_agent.py`) — With Self-Verification Loop

This is the most sophisticated agent. It implements a **generate-then-verify loop** to eliminate hallucinations.

```
Query
  |
  v
Hybrid Search (top_k=30) --> Reranker (top_k=5) --> MultiHop Expander
  |
  v
[Generation Attempt 1-3]
  |-- Gemini 3.7 Flash generates answer with citations
  |
  v
[Verification]
  |-- Gemini 3.5 Flash grades: is the answer FULLY supported by the con
  |-- If FAIL: retry generation (up to 3 times)
  |-- If PASS: return answer to user
```

**Why use a separate model for verification?**
A model often believes its own outputs. A *different model* (or different temperature setting) acts as a more objective judge. This is called **LLM-as-judge** and is a standard production pattern.

```
# qa_agent.py — The loop
```

```
for attempt in range(3):
    chain = self.generation_prompt | self.llm | StrOutputParser()
    answer = chain.invoke({"context": context_str, "question": question

    verify_chain = self.verification_prompt | self.verifier_llm | StrOu
    verification = verify_chain.invoke({"context": context_str, "answer

    if verification.strip().startswith("PASS"):
        return answer    # ✓ Verified, hallucination-free!
    else:
        continue         # ✗ Retry with a fresh generation
```

### 6.2 Review Agent (`agents/review_agent.py`)

Uses the first 500 characters of the diff as a search query (keyword-rich), retrieves related existing code, then asks Gemini to review the diff *in context of* the existing codebase. This catches breaking changes that a standard code review bot would miss.

### 6.3 Test Agent (`agents/test_agent.py`)

Retrieves the existing callers/callees of the target function, providing the LLM with the full picture of how the function is *actually used in production*. This leads to much better test cases (e.g., knowing the real input types and edge cases).

### 6.4 Docs Agent (`agents/docs_agent.py`)

Generates structured Markdown documentation. By providing dependency context, the docs include accurate descriptions of what the function calls internally, which is usually missing from standard LLM-generated docs.

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## 7. Phase 5: Enterprise Resilience (`utils/llm_factory.py`)

### 7.1 The Problem with Hardcoded LLM Instantiation

If you write `llm = ChatGoogleGenerativeAI(...)` directly in every agent, then:

1. **503 errors crash the entire pipeline** — no retry.
2. **Changing providers** requires editing 5+ files.
3. **No fallback** when Gemini's free tier rate-limits you.

### 7.2 The Factory Pattern Solution

We created a **centralized LLM factory** — a single function that all agents call. It handles all resilience logic in one place.

```
# utils/llm_factory.py
def get_llm(model_class: Literal["main", "fast"]):
    # "main" → Gemini 3.7 Flash / Claude 3.5 Sonnet (complex tasks)
    # "fast" → Gemini 3.5 Flash / Claude Haiku (routing, verification)
```

```
    primary_llm = ChatGoogleGenerativeAI(
        model=gemini_main,
        temperature=0,
        max_retries=3,    # Built-in exponential backoff!
        api_key=google_key
    )

    fallback_llm = ChatAnthropic(
        model=anthropic_main,
        temperature=0,
        max_retries=3,
        api_key=anthropic_key
    )

    # .with_fallbacks() is a LangChain LCEL method.
    # It catches exceptions from primary_llm and transparently retries
    return primary_llm.with_fallbacks([fallback_llm])
```

### 7.3 Resilience Flow Diagram

```
Agent calls: chain.invoke(...)
        |
        v
[Primary: Gemini 3.7 Flash]
  Attempt 1 → 503 Error!
  Wait 1s...
  Attempt 2 → 503 Error!
  Wait 2s...
  Attempt 3 → 503 Error!
        |
        | (all retries exhausted)
        v
[Fallback: Claude 3.5 Sonnet]
  Attempt 1 → SUCCESS ✓
        |
        v
Response returned to Agent
(User sees no error or interruption)
```

---

# 8. Phase 6: Evaluation Harness (RAGAS)

We don't just "eyeball" results. We implemented **mathematical evaluation** using the RAGAS framework.

**Three Core Metrics:**

1. **Context Precision**: Of all chunks retrieved, what fraction were actually relevant? (Measures retrieval accuracy)
2. **Context Recall**: Of all relevant chunks that *should* have been retrieved, what fraction were? (Measures retrieval coverage)
3. **Faithfulness**: Is every claim in the generated answer supported by the retrieved context? (Measures hallucinations — 1.0 = no hallucinations)

**Our benchmark dataset (`eval/benchmark.json`)** contains hand-crafted question/answer/context triples derived from the dummy repo. We feed these to RAGAS and get objective scores to compare different retrieval strategies.

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# 9. Phase 7: Web Dashboard (FastAPI + Custom UI)

## Backend: `api/main.py`

FastAPI exposes two endpoints:

- `GET /health` — Health check
- `POST /api/chat` — Accepts `{"message": "..."}`, invokes LangGraph supervisor, returns `{"response": "...", "intent": "..."}`

The static frontend files in `public/` are mounted directly onto the root URL, so the entire app is served from one process.

## Frontend: `public/`

Three files build the complete UI:

- `index.html` — Semantic HTML structure (header, chat window, input form)
- `styles.css` — Dark glassmorphism design (CSS variables, backdrop-filter, animations)
- `script.js` — Handles form submission, loading states, `fetch()` to `/api/chat`, and `marked.js` markdown rendering

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# 10. End-to-End Request Lifecycle

**User types: "What does the multiply function do?"**

```
[1] Browser → POST /api/chat {"message": "What does the multiply functi
[2] FastAPI → supervisor.graph.invoke({"input_text": "What does..."})
[3] LangGraph enters Supervisor node
[4] Supervisor: Gemini 3.5 Flash classifies → intent = "qa"
[5] LangGraph routes to QA node
[6] QA Agent: HybridSearcher.search("What does the multiply function do
    [6a] Dense: Qdrant returns top 30 semantic matches (cosine similari
    [6b] Sparse: BM25 returns top 30 keyword matches ("multiply" exact
    [6c] RRF merges & ranks → merged top 30 list
[7] QA Agent: Reranker.rerank(query, top_30, top_k=5)
    → Cross-Encoder scores each (query, chunk) pair jointly → top 5
[8] QA Agent: MultiHopExpander.expand(top_5)
    → Reads dependency_graph.json → adds callers/callees → ~8 chunks to
[9] QA Agent: format_context(chunks) → builds a formatted string with f
[10] QA Agent: generation_prompt | llm | StrOutputParser()
     → Gemini 3.7 Flash generates answer with citations
[11] QA Agent: verify_chain → Gemini 3.5 Flash grades → "PASS"
[12] LangGraph reaches END node
[13] FastAPI returns {"response": "The multiply function...", "intent":
```

```
[14] Browser: marked.parse(response) renders Markdown → hljs highlights
[15] Intent badge "qa" appears above the response bubble
```

---

# 11. Key Interview Questions & Answers

**Q: "Why did you use Tree-Sitter instead of LangChain's standard text splitters?"**

> LangChain's text splitters (character, recursive character) split by character count or newlines. For code, this frequently breaks functions in half, giving the LLM a fragment with no context. Tree-Sitter parses the actual syntax of the language, guaranteeing we always extract complete, syntactically valid code blocks. This directly improves context precision.

**Q: "Explain Reciprocal Rank Fusion. Why is it better than just taking the top results from each search?"**

> Simply concatenating both lists gives equal weight to rank #1 and rank #30. RRF uses `1/(k+rank)` to assign scores, so high ranks get significantly more credit. A chunk appearing at rank #2 in both lists is almost certainly more relevant than one appearing at #1 in only one list. RRF elegantly handles this without requiring any model training.

**Q: "Why not just use a larger context window and dump the whole codebase in?"**

> Three reasons: (1) Cost — LLM APIs charge per token. A 1M-token prompt is extremely expensive per query. (2) Quality — Research shows LLMs have an "attention sink" problem where they focus on the beginning and end of long contexts, ignoring the middle. (3) Latency — A massive context takes much longer to process. RAG solves all three by providing only the minimal necessary context.

**Q: "How do you prevent hallucinations?"**

> We use a generate-then-verify loop in the QA Agent. After generation, a second LLM call acts as a judge and checks if every claim in the answer is factually supported by the retrieved context. If not, it returns "FAIL" with a reason and we retry. Additionally, the system prompt explicitly instructs the generator to ONLY use information from the provided context and to cite file paths and line numbers.

**Q: "How does the multi-provider fallback actually work in LangChain?"**

> LangChain's LCEL (LangChain Expression Language) provides a `.with_fallbacks()` method on any Runnable. When an exception matching `exceptions_to_handle` is raised (503, 429, etc.), LangChain catches it internally and re-runs the same chain against the fallback model. The caller (our agent code) sees a successful result and never knows a failover occurred. The `max_retries` parameter on each model adds exponential backoff before falling over.